Edge Colors on the Copies Graph

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The graph. Every node that sits in two or more subgraphs is replaced by one copy per subgraph. Each copy keeps all of the node's edges. A *coloring* sorts the edges into groups: inside edges (both ends in one subgraph), outside edges, sibling edges (between two copies of one node), and our clique. Each group gets its own weights in the GNN.

What we ran, in four steps, all on GLASS's splits:

1. 23 colorings, each with our training and with GLASS's training, 3 seeds. Two colorings have no copies: the plain graph, and the plain graph split into inside/outside edges.
2. The 3 best colorings $\times$ 4 aggregations $\times$ 7 poolings, 3 seeds.
3. The 3 best setups $\times$ a small grid (hidden size, lr, dropout, edge counts), 3 seeds.
4. The 3 finalists and our previous best model, 10 seeds.

em_user, hpo_metab and hpo_neuro ran without copies only. With copies, their graphs have 35M to 1.3B edges and do not fit in GPU memory.

	ours		published	
dataset	new best	previous best	GLASS	GPEN
ppi_bp	59.9 ±0.9	56.3 ±2.8	61.9 ±2.2	**64.4** ±0.9
hpo_metab	61.1 ±2.3	60.2 ±2.7	61.4 ±1.6	**63.9** ±0.9
hpo_neuro	**69.6** ±0.9	68.7 ±0.9	68.5 ±1.6	69.1 ±0.6
em_user	73.1 ±6.0	80.8 ±8.8	88.8 ±1.9	**91.2** ±1.3
density	94.8 ±3.2	95.1 ±2.8	93.0 ±2.8	**95.6** ±1.6
cut_ratio	89.1 ±4.0	**93.7** ±2.8	93.5 ±1.9	93.6 ±1.6
coreness	86.5 ±5.4	83.5 ±6.8	84.0 ±2.8	**87.6** ±1.9
component	**100.0** ±0.0	99.7 ±0.6	**100.0** ±0.0	**100.0** ±0.0

Table 1: Test micro-F1 (%), mean ± std over 10 seeds. New best: the step-4 finalist with the highest validation score. Previous best: our best model per dataset before this run, rerun on the same splits. GLASS and GPEN: published numbers from the GPEN paper. Bold: the highest score in the row.

dataset	coloring	training	aggregation	pooling
ppi_bp	inside / outside, no copies	ours	mean	sum / $\sqrt{size}$
hpo_metab	plain graph	GLASS	sum	gated attention
hpo_neuro	inside / outside, no copies	GLASS	attention	multistat
em_user	plain graph	GLASS	GCN	max
density	inside / outside-in / outside-out, copies	GLASS	sum	mean
cut_ratio	4 groups*, copies	ours	sum	mean
coreness	copies graph + clique	GLASS	attention	gated attention
component	plain graph	GLASS	attention	multistat

Table 2: The setup behind each "new best" number. Outside-in / outside-out: outside edges split by whether the receiving node is in a subgraph. *Inside edges split by whether they touch a copy; outside edges split by whether both ends are in subgraphs.

Findings

- **Better on 5 of 8 datasets.** The largest gains are on ppi_bp (+3.6) and coreness (+3.0). hpo_neuro (69.6) is above both GLASS (68.5) and GPEN (69.1), and hpo_metab (61.1) is 0.3 below it.
- **Worse on cut_ratio (-4.6) and em_user (-7.7).** em_user varies a lot between seeds (± 6 to ± 9).
- **Copies help only on the synthetic datasets.** In step 1, test with no copies $\rightarrow$ best coloring with copies (3 seeds): density 69.8 $\rightarrow$ 92.6, cut_ratio 80.4 $\rightarrow$ 90.0. On ppi_bp, coreness and component there was no gain.
- **Sibling edges do nothing.** With or without them, results differ by less than 1 point on every dataset.
- **GLASS's training wins on most datasets.** 6 of the 8 new best setups use it. In step 1 it was ahead by 8 points on coreness and 4 on density (validation).
- **Aggregation and pooling mattered more than tuning.** Step 2 raised hpo_metab's validation score from 62 to 67. Step 3's grid added at most 2.5 points.
- **Validation does not predict test well.** On hpo_metab the new best scores 69 on validation but 61 on test. On cut_ratio and density the previous model had the higher validation score (97.9 vs 95.0 and 97.9 vs 97.3), so picking by validation over all four models would have kept the previous one there.